



ME 300: Engineering Thermodynamics

Example: Brayton Cycle

Example

A gas turbine running an ideal Brayton cycle has a steady mass flow of 30 kg/s. The inlet temperature is 300 K and the max temperature is 1050 K. If the pressure ratio of the cycle is 8.5, determine the thermal efficiency and the net power produced.

Given: $\dot{m} = 30 \text{ kg/s}$

$$T_1 = 300 \text{ K}$$

$$T_{\max} = 1050 \text{ K}$$

$$P_2/P_1 = 8.5$$

Assume: air standard

: ideal $\rightarrow$ reversible

: steady-flow

: quasi-steady

Find: η_{th}

$$\dot{W}_{net}$$

Efficiency

$$\eta_{th} = 1 - OPR^{\frac{1-\gamma}{\gamma}}$$

$$= 1 - (8.5)^{\frac{0.4}{1.4}}$$

$$\gamma = 1.4$$

$$\eta_{th} = 0.46$$



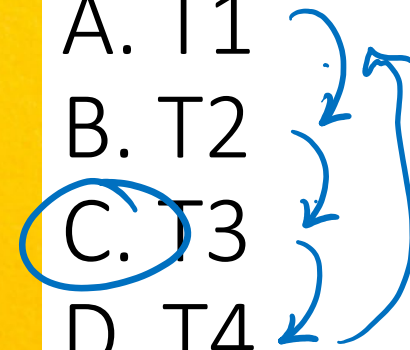
What is the highest temperature in the cycle?

A. T1

B. T2

C. T3

D. T4



Thermodynamic states

<u>State</u>	<u>P</u>	<u>T [K]</u>
1	P_1	300
2	$8.5 P_1$	553
3	$8.5 P_1$	1050
4	P_1	570

$$\Delta S = 0 \quad T_1 P_1^{\frac{1-\gamma}{\gamma}} = \underline{T_2 P_2^{\frac{1-\gamma}{\gamma}}}$$

$$\Delta S = 0 \quad T_3 P_3^{\frac{1-\gamma}{\gamma}} = T_4 P_4^{\frac{1-\gamma}{\gamma}}$$

$$C_p = 1001 \text{ J/kg}\cdot\text{K}$$

Work

$${}_1\dot{W}_2 = -\dot{m}(h_2 - h_1) = -\dot{m}c_p(T_2 - T_1) = -30(1001)(553 - 300) = -7.5 \text{ MW}$$

$${}_3\dot{W}_4 = -\dot{m}c_p(T_4 - T_3) = -(30)(1001)(570 - 1050) = 14.4 \text{ MW}$$

$$\dot{W}_{\text{net}} = {}_1\dot{W}_2 + {}_3\dot{W}_4 = -7.5 + 14.4 = > \boxed{6.8 \text{ MW}}$$